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Characterization of Recycled Dredged Sediments: Toward Circular Economy in Road Construction

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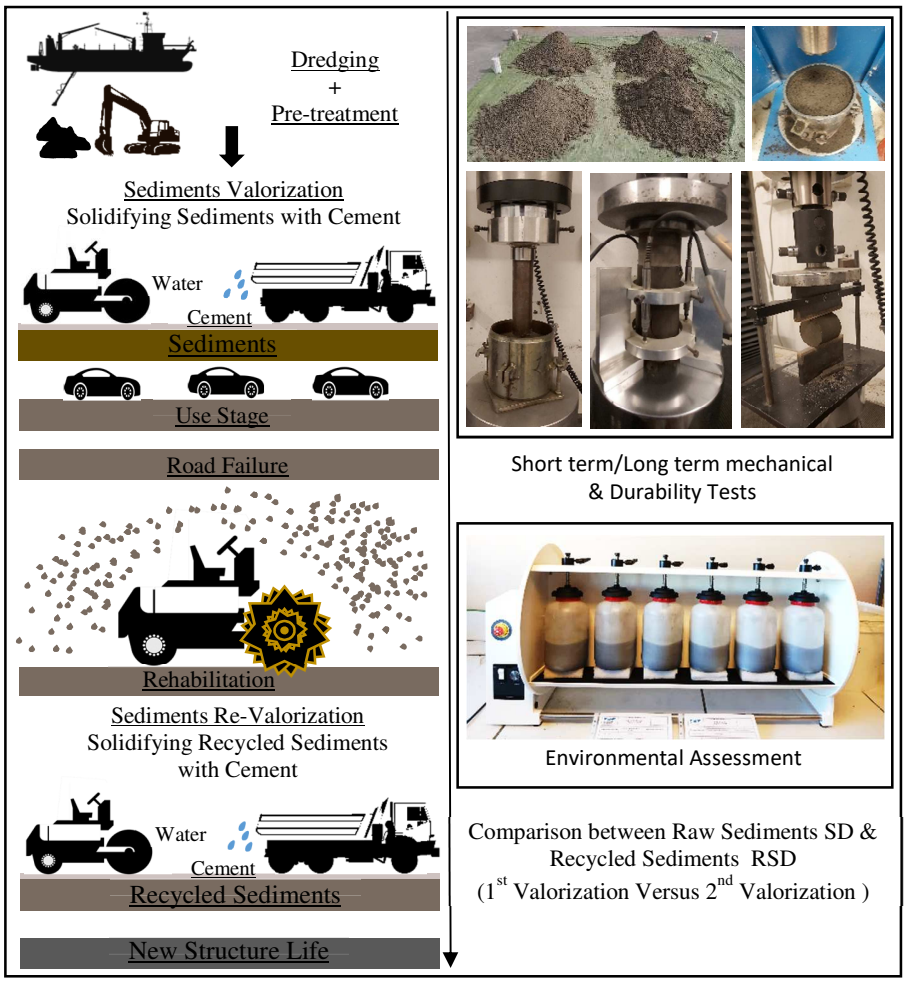
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Abstract

Granular solid materials are essential for civil engineering projects, but their limited availability and environmental impacts have resulted in exploring alternatives aligned with sustainable development. Utilizing waste materials as dredged sediments offer promising solutions for more sustainable construction practices, especially in road construction. However, end-of-life management of these materials remains a challenge, requiring a circular approach to ensure their sustainable use. This research aims to evaluate the viability of revalorization of recycled sediments (RSD) from a demolished road after its failure for their second use in road structures compared to the first valorization of raw sediments (SD). The usability and environmental acceptability were evaluated through physical, mechanical, durability, and environmental testing. The physical properties of RSD show lower particle density (2.55 g/cm^3) than SD (2.59 g/cm^3), and have less organic matter content and a non-plastic behavior. The immediate California bearing ratio (I-CBR) after solidification/stabilization treatment was around 165% higher for RSD than SD. The mechanical and durability performances of RSD were also better than SD in terms of water sensitivity and frost resistance, as well as at different curing ages 7, 28, 60, and 90 days. Additionally, the environmental leaching behavior of the trace elements in RSD matrix before and after treatment differed from SD, indicating varying road uses' applicability. The findings highlight the potential of utilizing various wastes in constructions, thus promoting a closed-loop circular approach and reducing the need for virgin resources.

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Graphical Abstract

Keywords: Dredged sediments, Revalorization, Circular Economy, Durability, Road Rehabilitation, End of life Management, Life Cycle Assessment

1. Introduction

Granular solid materials such as gravel and sand are essential components in civil engineering projects. However, the availability or access to these resources is limited. Moreover, their extraction can have significant environmental impacts. Indeed, the global consumption of sand and gravel in the world is estimated to be around 40-50 billion tons per year and this amount is expected to increase due to demographic increase, rapid urbanization, and infrastructure development [1]. As a result, there has been a greater emphasis on exploring alternatives that align with the principles of sustainable development, which has become the main concern of societies' development targets in the current century [2]. In the construction sector, sustainable alternatives include maximizing construction durability, designing easily rehabilitable structures, and increasing efficient and safely waste incorporation.

Utilizing waste materials like bottom ash, fly ash, rice husk ash, slag, rubber, construction and demolition waste (CDW), and sediments[3], as well as incorporating innovative applications and materials like pervious concrete, self-healing concrete, fibers, polymers, and others offer promising solutions for more sustainable construction practices[4–6]. In fact, valorizing diverse wastes and minimizing solid waste consequently contributes to over one–third of sustainable development goals initiated by the united nations[7]. Utilizing these materials has emerged as a promising solution to alleviate the depletion of mineral solids and provide a cost-effective alternative for civil engineering applications. Subsequently, road construction stands out as a key application area where these materials can be utilized. In recent years, dredged marine sediments have gained attention as potential alternative materials for different road applications [8–14].

Despite these benefits, there are logistical and environmental challenges associated with the use of waste materials, including concerns about their long-term management. Especially in the case of dredged sediments, which have in France an estimated potential annual volume of 40 to 50 million cubic meters [15].

Due to the rigorous demands of road structures, alternative materials must possess specific mechanical and durability properties to ensure long-term performance and adhere to certain environmental thresholds. The material's mechanical and durability properties, including immediate California bearing ratio (I-CBR), unconfined compressive strength (UCS), indirect tensile strength (ITS), modulus of elasticity (E), water sensitivity, freeze-thaw (FT) resistance, and volumetric swelling (VS) must be carefully evaluated. These properties ensure that the material withstands the demands of its intended road application and they conform to environmental guidelines such as those outlined in the SETRA 2011 methodological guide [16] used in France.

Based on previous studies, dredged marine sediments have been treated in various ways to enhance their mechanical and environmental leaching behavior, making them suitable for higher-performance road layers or as backfill material [17]. The sediments have been mixed with virgin or waste-based granular correctors[13,18], or further stabilized/solidified using lime and cement [19], by-products[11], or hydraulic road binder [20] to upgrade their mechanical properties for use in the

capping layer, subbase, or base course layer applications. Technical acceptability has been synchronized with environmental acceptability, as demonstrated in Mkahal et al. 2021 [8], which assessed sediment co-valorization with other wastes in different pH conditions for use as backfill material. Nevertheless, none of these studies have considered or addressed the fate of the material after the end of its road life cycle, highlighting a gap in research on the long-term use of these materials in road construction.

The concept of circular economy has further fueled this interest, as it promotes moving waste materials into useful resources through a sustainable approach [21]. The traditional linear economy was first updated with the circular feedback loops which were originally initiated by Stahel in 1976 [22,23]. This began with the conventional 3R concept (Reduce, Reuse, and Recycle) [24,25]. It was then followed by the real circular economy approach based on the 6R concept, which combines the three aspects of sustainability: economic, environmental, and societal through the principles of Reduce, Reuse, Recycle, Recovery, Remanufacturing, and Redesign. [26–29]. To implement the circular economy concept, these principles can be categorized into three main strategies: closing loops, slowing loops, and narrowing loops [21,30]. In the construction sector, this involves creating circular material flow, increasing the lifetime of structures, and maximizing waste incorporation in construction.

While sediments were previously considered waste, recent regulations have provided criteria for their exclusion from waste classification under certain circumstances as in the order of June 4, 2021 [31]. To ensure the successful implementation of circular economy principles in road construction, it is important to consider the entire life cycle of sediments and their potential for reuse after initial valorization. Thus, studies on the long-term management of sediments are necessary to identify viable opportunities for their continued utilization and integration into the circular economy chain.

However, long-term management of these valorized materials remains a significant challenge, with many waste materials being landfilled [32] and exacerbating existing land use problems. To overcome this issue, it is essential to assess the usability of these materials and to explore the potential for their revalorization, which refers to the valorization of a material or structure for a second time after its initial use has ended. The revalorization of sediments, in particular, is a novel issue that requires attention from the initial design stage of incorporating raw sediments into road structures. By considering revalorization in the design phase, we can establish a more durable and circular approach to road construction that reduces waste and enhances sustainable resource use.

This study aims to evaluate the circularity and durability of demolished road materials made from dredged sediments for use in new road structures. Specifically, we compare the valorization of raw sediments (SD) versus recycled sediments (RSD) in terms of their mechanical and environmental performance. The results of this study have significant implications for the development of circular

129 solutions in road construction. To ensure the usability of recycled sediments, we conducted several
130 mechanical and durability tests following the revalorization process. Additionally, we evaluated the
131 leaching behavior of trace elements and salts in both SD and RSD before and after treatment to
132 verify their environmental acceptability for use in different road types.
133 The findings of this study have the potential to contribute to a more durable approach to road
134 construction by promoting the revalorization of waste materials and reducing virgin resource
135 consumption. As such, this research is of interest to researchers, practitioners, and policymakers
136 seeking to advance circular economy principles in the construction sector.
137
138 ## 2. Materials and Methods
139
140 ### 2.1. Studied material
141
142 The sediments used in this study were obtained from the maritime port of Dunkirk, located in
143 northern France. SD materials are classified as non-submersible due to their content of pollutants,
144 including trace amounts of both metallic and organic compounds [33,34]. Prior to their first use, the
145 sediments were subjected to a dehydration process in specialized basins, which effectively reduced
146 their high water content [35]. Subsequently, the sediments were transported to the laboratory site in
147 large bags and homogenized before being quartered to obtain representative samples in accordance
148 with the standard NF ISO 11464 [36]. Geotechnical experiments were conducted after oven-drying
149 the sediments at 40°C when necessary. To solidify the sediments, ordinary Portland cement (OPC)
150 CEM I 52.5 R supplied by Lafarge Holcim was used as a binder. The cement contains 59.8% CaO,
151 17.9% SiO_2, 2.6% Fe_2O_3, 4.4% Al_2O_3, 0.9% MgO and other components.
152
153 ### 2.2. Physio-chemical and Mechanical Tests
154
155 In order to assess the physio-chemical properties of sediments, several key parameters were
156 evaluated based on established European and French test standards. These included the initial water
157 content [37], organic matter content [38], loss on ignition [39], particle density [40], grain size
158 distribution[41], Atterberg limits [42], methylene blue value (MBV) measurement [43], and sand
159 friability [44]. From the measured characteristics, the materials were classified according to NF P11-
160 300 French classification [45] as well as European Soil Classification System (ESCS) [46] . X-Ray
161 Diffraction (XRD) and mercury intrusion porosimetry (MIP) tests were used to analyze the materials'
162 crystalline-phase-mineralogy and microstructural properties.
163
164 The mechanical properties of the material were evaluated using established testing procedures.
165 Maximum Dry Density (MDD) values were determined through the modified proctor test, in which
166 an automatic Proctor machine was used to compact the material in CBR molds, as per NF EN 13286–
167 2 standard [47]. The compacted material's I-CBR was determined according to NF EN 13286–47
168 [48] using the Instron 5500R - 150 kN Mechanical press.

The UCS, ITS, and E tests were conducted using Instron machine and in accordance with the standards NF EN 13286-41[49], NF EN 13286-42[50], and NF EN 13286-43 [51] respectively. Cylindrical specimens were prepared for the UCS, ITS, and E tests by compacting the material using axial static compression, as specified in the NF EN 13286–53 standard[52]. They were prepared at their respective MDD with a design factor of 98.5% for SD and 97% for RSD, and at their optimum water content (OWC), according to the standard NF P98-114-3 [53]. Three replicate tests were conducted to ensure the reliability of the results, and the error bars in the graphs below represent the standard error of the mean.

Specimens, with a 50 mm diameter and 100 mm height, were used for UCS and E tests, while specimens with a diameter of 50 mm and a height of 50 mm were used to perform the ITS test. Tap water was used in the preparation of mixtures. The specimens were manufactured and stored in their sealed cases in a curing room of relative humidity (RH) %=98% and constant temperature (T)= 20°± 2 °C until the test realization date (after 3,7,28,60, and 90 days). **Fig. 1** shows a schematic representation of the specimens undergoing UCS, ITS, and E tests on their realization day. These methods are well-established and widely accepted within the field of road materials, and the resulting data provide important insights into the material's suitability for various applications in road construction.

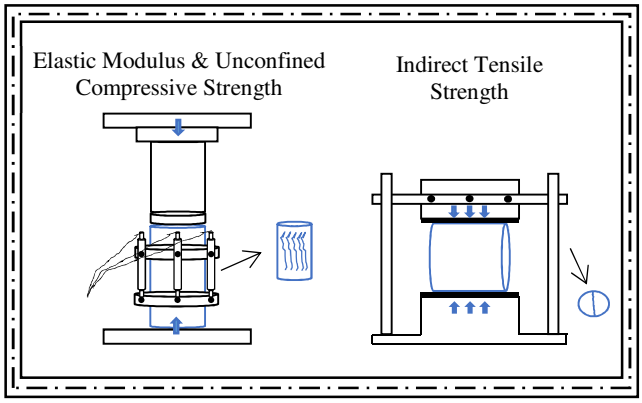


Fig. 1 Specimens Undergoing UCS, ITS, and E Tests on Realization Day

2.3. Durability Tests

Water sensitivity tests are a valuable tool for investigating the effects of complex wet weather or rainy conditions on samples, where substantial amounts of water infiltrate the samples. In this study, the water immersion test was performed in accordance with the NF EN 14227-15 standard [54]. Following a 28-day curing period at a room temperature of 20°C, cylindrical specimens were immersed in deionized water for 32 days under identical temperature conditions[54]. Subsequently, UCS tests were performed on the samples and the results were compared with those of the control specimens that were cured for 60 days under normal conditions.

The response of the materials to thawing-freezing processes was assessed using UCS measurements after subjecting cylindrical specimens to 20 freezing-thawing cycles, in accordance with the NF P

98-234-1 test standard [55]. Two types of compression and two different dimensions were used to manufacture the specimens. Vibro compression (VC) [56] was employed to create specimens with a diameter of 100 mm and height of 200 mm, while static axial compression (SC) [52] was used to produce specimens with a diameter of 50 mm and a height of 100 mm. Prior to testing, the specimens were stored at 20 °C for 28 days. Subsequently, half of the specimens were subjected to 20 freezing-thawing cycles in a climatic chamber, while the remaining half were used as control specimens and tested at 28 days. The freezing-thawing cycle consisted of a 4-hour period at +10 °C, followed by a 4-hour period to decrease the temperature from +10 °C to -10 °C, 14 hours at -10 °C, and a final 2-hour period to increase the temperature from -10 °C to +10 °C. This testing protocol provides important information on the durability and performance of the material under harsh environmental conditions.

To assess the suitability of materials treatment, the accelerated volumetric swelling (VS) and ITS were evaluated. To accelerate the hydraulic setting of the specimens, the procedures outlined in the NF P94-100 [57] and NF EN 13286-49 [58] standards were employed. Accordingly, the specimens were stored for 4 hours at a temperature of 20°C and RH of 98% before immersion in water at a temperature of 40°C for 7 days. The material VS was subsequently determined with respect to the initial volume of the material. The suitability of the materials for treatment was then evaluated based on the VS and ITS results after acceleration procedures. Materials' treatment was considered suitable if they exhibited a VS of less than or equal to 5% and an ITS of greater than or equal to 0.2 MPa [57].

2.4. Environmental assessment

A one-stage batch equilibrium leaching experiment was conducted to determine the leaching concentrations of different inorganic elements in leachates at a liquid-to-solid ratio (L/S=10). The leaching procedure involved liquid-solid separation via filtration with a 0.045 µm pore size filter in accordance with the NF EN 12457-2 standard [59]. Inductively coupled plasma-optical emission spectrometry (ICP-OES) and ionic chromatography were used for elute analysis of trace elements and anions, including Cr, Ba, Cd, Zn, Cu, Pb, Ni, Sb, Se, Mo, As, fluoride, chloride, and sulfate. Quantification Limits (QL) of 0.4, 0.8, 0.9, 1, 2.1, 3.2, 4.7, 5.7, 8.3, 8.8, 11, 100, 1000, and 1000 (µg/L) were used to determine the concentration of each element, respectively. The experiment was carried out following accepted scientific protocols, and the resulting data is presented herein.

2.5. Manufacturing RSD and the Revalorization Scenario

Fig. 2 shows the scheme that represents the way of RSD manufacturing with the scenarios of comparison between SD and RSD. To produce a representative material of the recycled sediments status, the raw dredged sediments were treated with 6% of ordinary Portland cement and then compacted to 98.5% (design factor) [60] of the mix's MDD using the modified Proctor energy.

During compaction, the water content was adjusted to be at the mix's OWC. The material was immediately placed inside a curing room with (T=20°C, RH= 98%) under sealed conditions for seven days to represent real-life construction companies' actions for water saturation for these initial days. Then for the remaining period, the materials are kept outside until the age of 90 days. At this age, the materials give an insight into the long-term behavior on the mechanical levels [60]. Then the materials were crushed at a 6mm opening crusher and thus RSD is obtained. This opening is used to describe the effect of the road reclaiming machines. Therefore, RSD stands for crushed road materials based on a mix of sediments and OPC. Which is then treated as a new raw material resource. RSD requires a treatment to be suitable as a subbase material, as in the case of SD without treatment. Accordingly, the mechanical behavior of SD and RSD after treatment with 6% of OPC and among different curing periods(from 3 up to 90 days) were addressed. The selection of the 6% treatment was based on its demonstrated effectiveness in a range of sediment valorization studies, as well as its successful implementation in both laboratory and on-site applications, as evidenced by previous research [10,12,61–63]. In parallel, materials' durability and environmental impacts are assessed for their 1st and 2nd valorization (revalorization) as subbase materials. Hereafter, SD and RSD are termed as initial materials status, while SD-6C and RSD-6C are denoted as treated materials.

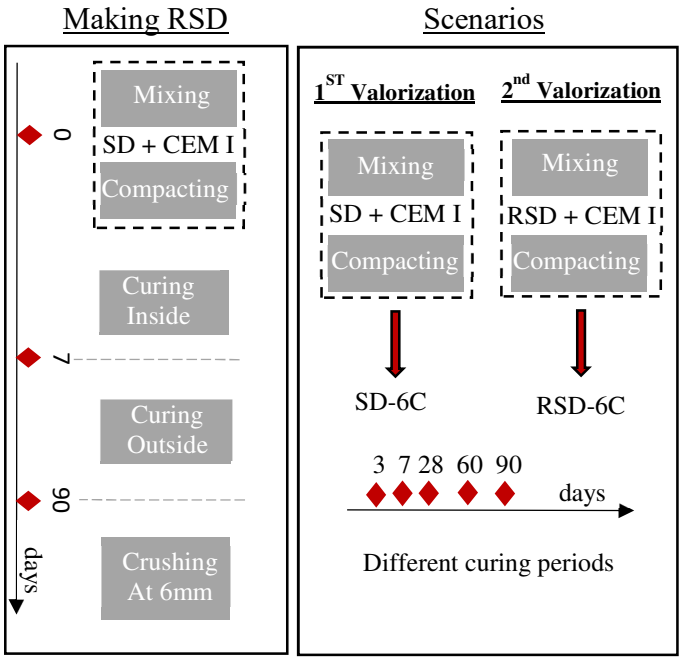


Fig. 2 Manufacturing RSD and valorization scenarios

3. Results

3.1. SD & RSD Physiochemical Properties

The physicochemical and mineralogical characterization of SD and RSD are shown in **Table 1.** The evolution of sediments upon recycling is mainly observed in material plasticity and granulometry.

As reported by Wang 2011, the solidification process decreases the sediment's plasticity and reduces organic matter, and that's because of the hydration reaction and its formed products in the sediment matrix [64]. Moreover, RSD has a lower particle density (2.55 g/cm^3) than SD (2.59 g/cm^3) due to the presence of recycled cement in the sediment matrix, which has a lower density than the initial cement status. Recycled cement paste has a density of 2.4 g/cm^3, whereas standard Portland cement has a density of 3.1 g/cm^3. The results of organic matter, methylene blue value, and particle density of SD are consistent with the previous research conducted by Maherzi et al. 2018 [9], which dealt with marine sediments dredged from the Brest harbor area for road construction and had similar mineralogical composition. However the physiochemical results of RSD were more similar to those of CDW material used in the study of Mkahal et al. 2021[8], which showed a low methylene value (0.3g/100g) with no plasticity, lower particle density of 2.5 g/cm^3, and presence of portlandite phase in its mineralogy.

Table 1 Raw and recycled sediments' physiochemical and mineralogical properties

Parameter	SD	RSD
Water content (%)105°C (24h)	21.69	14.5
Organic matter (%) at 450°C	5.84	4.53
Loss on ignition (%) at 550°C	7.34	6.14
Specific gravity (g/cm^3)	2.59	2.56
Methylene blue-value (g/100g)	1.15	0.51
Plastic limit (PL)	24.96	
Liquid limit (LL)	39.5	Non Plastic
Plasticity index (PI)	14.55	
Sand friability coefficient (Fs)	NA	35.28
Grains ≤ 2µm (%)	14.47	2.04
2µm < Grains < 63µm (%)	74.5	13.88
Grains ≥ 63 µm (%)	11.03	84.08
ESCS Classification	Silt with medium plasticity	Sand with clay or silt
NF P11-300 Classification	Silt with Low plasticity (A1)	Sandy soil with fines (B2)
Mineralogy (XRD)	Quartz, Muscovite,Calcite Pyrite,Rutile	SD phases, Ettringite, Portlandite

3.2. Compaction and Immediate California Bearing Index

Fig. 3 (a) shows the proctor compactions curves for SD and RSD. The optimum compaction level is determined by following the variation of MDD and water content of the material, ensured by the restriction between the two curves of saturation Sr=80% and Sr=100% as defined for SD. A bell-shaped curve is observed for SD and RSD initial and treated status. For SD, the initial raw status has an MDD of 1.79 g/cm^3 with 14.6 % as an OWC. After treatment, SD-6C has a slightly higher MDD and almost the same OWC (MDD=1.81 g/cm^3 and OWC =14.7 %). This slight increase in water demand can be ascribed to the hydration reaction. The MDD obtained for SD (1.81 g/cm^3) was slightly higher than the MDD of Wang et al. 2012 [19] findings (1.74 g/cm3) for silty sediments from Dunkirk area, which typically have higher plasticity. In the case of RSD, the initial MDD (1.7 g/cm^3) was lower than that observed for SD. However, its OWC (16.4%) is higher than SD's OWC. The decrease in MDD is attributed to the properties of the grains that make up RSD. Although the

material appears coarse based on the grain size distribution, it is actually made up of porous particles formed from finer particles that have a high water absorption due to the adhered cement paste [65]. The results of RSD material in this study are consistent with the findings of Cabalar et al. 2019 [66], where the incorporation of recycled concrete aggregates led to a decrease in the MDD of a clay and recycled concrete aggregates mixture, in comparison to a mixture of natural aggregates with clay. Moreover, RSD-6C has a higher MDD and OWC than the initial RSD values with 1.73 g/cm3 and 17 % respectively. Water demand increase can be attributed in part to the hydration reaction between the binder and the initial state of the materials, but it is primarily due to the porous nature of the particles that make up RSD. The pattern of OWC increasing after treatment for both materials is concordant with the results of a study by Zentar et al. 2021[63], which investigated the solidification/stabilization of dredged sediments by two types of cement, ordinary Portland cement and calcium sulfo-aluminate cement, for their use as a subbase road material.

The I-CBR indexes for these materials are illustrated in **Fig. 3 (b)**. For both materials, the I-CBR index increases from low values at a low water content to reach maximum I-CBR, and then it decreases to reach very small values (between 0% and 10%). The I-CBR at OWC is generally at the inflection point of the downward cure representing I-CBR index. From **Fig. 3 (b),** it can be inferred that the I-CBR index value for SD is 14%, while RSD has an I-CBR index value of 80%. The I-CBR value of SD at its OWC was similar to that found by Zentar et al. 2021 [63](15%). The I-CBR increase for RSD is mainly owed to the high strength of the recycled hydrated cement binds between the sediment particles in the RSD material. As the RSD constitutes one of the CDWs that usually have a high CBR value (>45%) [67–69] thus it can have comparable properties with such material. The sandy classified RSD material has a similar I-CBR value to sediments of the same origin that are mixed with dredged sand as granular correction in the study of Dubois 2006 [70], where the I-CBR value was greater than 80%. For the treated materials, both SD-6C and RSD-6C show an increase in the I-CBR indexes (55% and 90% respectively). The observed increase in strength can be ascribed to the formation of various cementitious products as a result of treatment with OPC, a finding that has been reported in several solidification/stabilization treatments of sediments using OPC [10,63,71]. RSD exhibits an elevated I-CBR index, which can be ascribed to the superior resistance of the formed aggregates during the initial valorization process. This is reflected in the sand friability coefficient of 35.28, further indicating the enhanced durability of the artificial aggregates RSD. According to the French earthwork classification [45], SD is classified as (A1) and RSD is classified as (B2). Hence, both materials have an I-CBR index higher than the backfill requirements for these two classifications I-CBR> 8% and I-CBR> 12% for SD and RSD respectively [72]. While for the road structure, the material can be classified into three levels: insufficiently stable materials (I-CBR < 25%), subbase materials (I-CBR> 35 %), and base course materials (I-CBR> 50 %) [73,74]. Consequently, SD-6C, RSD, and RSD-6C have attained the required I-CBR for subbase and base course layers. Of course, the materials have additional parametric studies to address their acceptability in these layers. Where I-CBR index is used to assess the material short-term capability of supporting construction machines' loads [48].

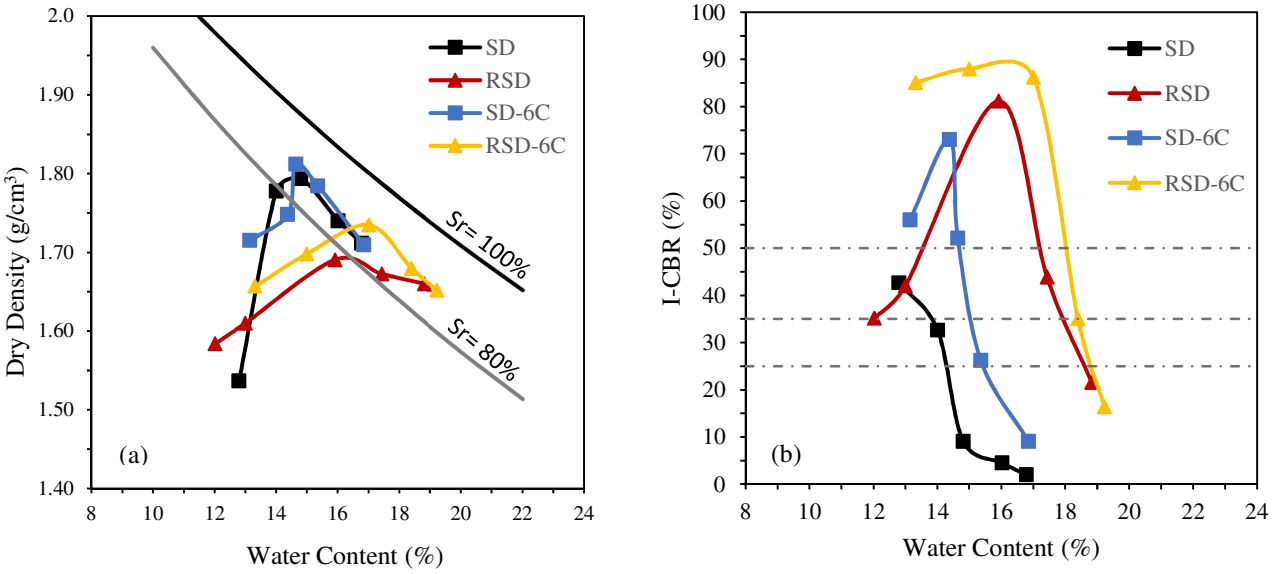


Fig. 3 (a) Compaction curves & (b) I-CBR Curves

3.3. Mechanical Performance

Road materials require certain mechanical properties in order to be used in the subbase or base course layer. These properties are aligned with the material construction age and also material exposure conditions. The age at which construction site traffic is permitted is dependable on the material UCS, and the material ITS is the conditional property for evaluating material at the first occurrence of frost. According to previous studies [13,63,75], solidification with cement improves sediments and CDWs' mechanical properties.

3.3.1. Unconfined compressive strength (UCS)

Materials UCS along the different curing ages 3,7,28,60 and 90 days are shown in **Fig. 4 (a)**. The initial materials SD and RSD show an unchanged in their UCS in the different curing periods, which is equal to 0.18 and 0.33 MPa respectively. However, SD-6C and RSD-6C show a large increase in their USC as the curing age increase. For SD-6C, the UCS increased from 1.81 MPa at the age of 3 days and keep increasing up to 3.89 MPa at 90d. Meanwhile, RSD at all curing ages has displayed a higher UCS than SD starting from 2.55MPa at 3d and up to 5.08MPa at 90d. Several studies have shown that CDWs have a UCS range between 5 and 7 MPa in the curing ages ≤ 28d [76–79]. In these studies, the CDWs' fractions and the binder type/content are the factors that affect CDWs' UCS. RSD-6C exhibits similar mechanical properties as CDWs, leading to its higher UCS compared to SD-6C.

Loudini et al. 2020 [80] reported that the strength measurements of SD solidified with 7% cement were significantly higher (1.54-2.47 times) than the UCS values at 7 days of curing. In this study, UCS values for SD-6C and RSD-6C at 28, 60, and 90 days of curing were also found to be higher (1.4-1.56 times for SD-6C, 1.19-1.64 times for RSD-6C) compared to UCS values at 7 days of

curing. The small difference between the UCS at 7 days and at the different curing ages in this study is attributed to the use of rapid cement CEM I 52.5R, which facilitates rapid solidification and is particularly suitable for road construction, which requires rapid strength development to support traffic and construction machinery. Construction traffic is determined by number of vehicles ≥ 3.5 tons, known as PL. Three levels of mechanical stress aggressiveness exist A (<20 PLs), B (20-500 PLs), and C (>500 PLs)[60]. The age at which the construction site traffic is permitted depends on the material UCS and traffic aggressiveness level. SD-6C and RSD-6C exceed the required UCS for all aggressiveness levels (A:1MPa, B:1.2MPa, C:1.5MPa) after 3 days of curing.

3.3.2. Indirect Tensile Strength (ITS)

Fig. 4 (b) presents the ITS of different materials during the curing ages 3, 7, 28, 60, and 90 days. SD and RSD materials show no change in their ITS values over the curing period, remaining at 0.025 and 0.056 MPa respectively. On the other hand, SD-6C and RSD-6C display a noticeable increase in their ITS as the curing age increases. The ITS of SD-6C rises from 0.26 MPa at 3 days to 0.5 MPa at 60 days. RSD consistently displays a higher ITS compared to SD in these curing periods, starting at 0.41 MPa at 3 days and reaching 0.64 MPa at 60 days. Then both materials increase to reach almost the same ITS at 90 days with 0.65 MPa. This trend is in agreement with the findings of Maherzi et al. 2018[9], who reported a similar increase in ITS values for SD treated with 6% cement over 90 days of curing for use as a capping layer.

Das and Sobhan 2013[81] classified particle shape into three categories: bulky, flaky, and needle-shaped particles, and noted that each shape has a significant impact on the physical and mechanical properties of the material [81]. Aggregates with higher angularity and rougher surface textures form a stronger mechanical bond with cement paste by interlocking, as reported by Kaplan's 1959 [82]. Consequently, the rougher texture material RSD, resulting from cement solidification leads to higher UCS and ITS values. However, the similarity in ITS between SD and RSD at 90 days of curing suggests that the physical and chemical bonds between the binder and particles in both materials have reached a state of maturity. As the mixture continues to cure, the bonds become stronger and more homogeneous, leading to a similar ITS for both materials. The cement-material bond strength may be stronger than the strength of the material itself [83]. This is another cause that may lead to these differences between SD and RSD, which depends on the complete hydration of cement over time. This demonstrates the importance of adequate curing time, which allows for the development of strong and stable mechanical and chemical bonds in the road material. The ITS of the materials SD-6C and RSD-6C at various curing stages are greater than 0.25 MPa, which is considered to be the minimum strength requirement for the first potential occurrence of frost [60]. This requirement is further discussed in relation to the impact of FT cycles in the FT section below.

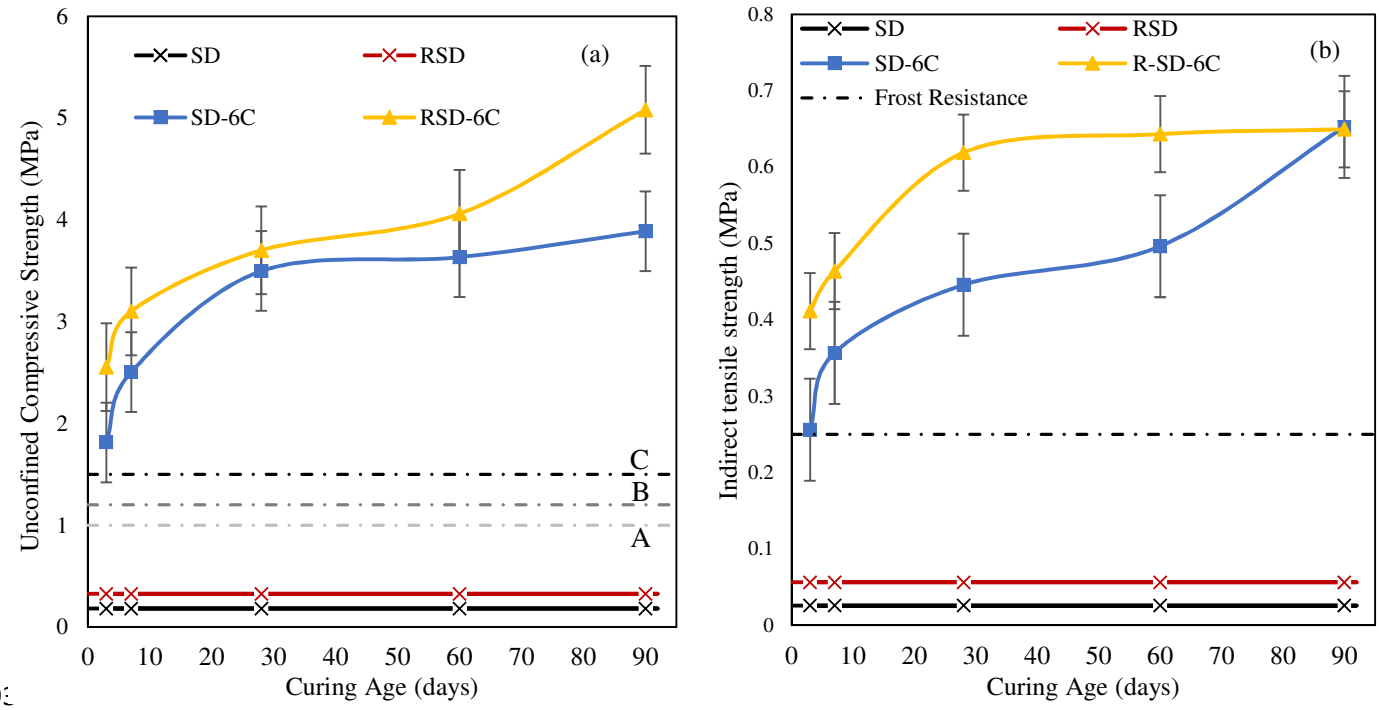


Fig. 4 (a) Unconfined compressive strength (b) Indirect tensile strength

3.3.3. Modulus of Elasticity (E)

The elastic modulus of a material is a fundamental mechanical property that describes its resistance to deformation under applied stress. To determine this property for hydraulically bound mixtures, the compression test method specified in the standard EN 13286-43 [51] is used. Specifically, the elastic modulus E of the cylindrical samples of diameter (D) is calculated based on the maximum force sustained (F_{Max}) at the maximum stress (σ Max) during the test, and on the corresponding strain value at stress equal to 30% of σ Max (ε 30%), as defined by equation (1) and shown in **Fig. 5**. The stress-strain curve has two regions: elastic and plastic. The elastic region is reversible and characterized by the material's stiffness, described by its elastic modulus. However, in the plastic region, the material undergoes irreversible deformation and its behavior is determined by its yield strength and strain-hardening properties[61]. This approach is widely used to assess the mechanical behavior of materials, and it has been validated by numerous experimental studies [9,10,18,61,63].

$$E = \frac{1.2\ F_{Max}}{\pi\ D^2\ \varepsilon_{30\%}} \quad (1)$$

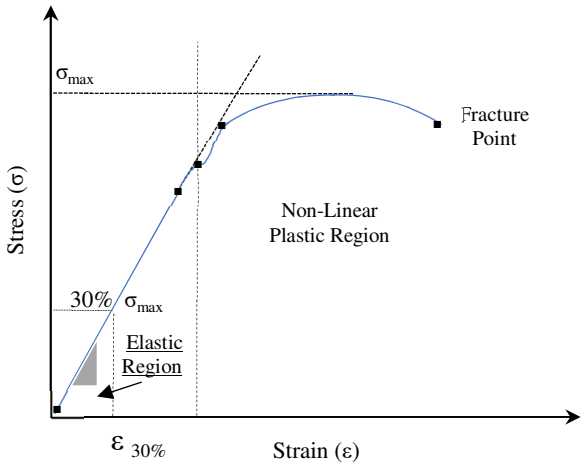


Fig. 5 Standard Stress-Strain Curve for Hydraulically Bound Mixtures

Fig. 6 compares the E of the two materials SD-6C and RSD-6C with respect to the curing time. The initial elasticity of SD and RSD was 0.3 GPa and 0.5 GPa, respectively, prior to curing. The results show that both treated materials SD-6C and RSD-6C have an increasing trend in elasticity as the curing time increases. At 3 days of curing, RSD-6C exhibited a higher elasticity compared to SD-6C, with a ratio of 2:1. This was due to RSD's larger particle size and initial higher strength. The difference in elasticity between RSD-6C and SD-6C decreased with the curing period increase, reaching a ratio of 1.4:1 after 90 days. Various studies have shown that the treatment of SD with different types of binders, such as OPC, calcium sulfoaluminate cement [63], a composite binder [84], fly ash [85], or basic oxygen furnace steel slag [11], can increase the sediment's stiffness and elasticity, resulting in appropriate properties for road construction. The elasticity of treated sediments increased with increasing curing time, indicating that longer curing periods result in greater elasticity.

The behavior of RSD differs from that of SD, as RSD exhibits stronger characteristics. It shares similarities with the results of sediment treated with sand as a granular corrector in previous studies [10,13]. For example, the elasticity results of RSD-6C at 28 days of curing are similar to those reported by Bourabah and Taibi 2018 [86] which range between 2.3 and 2.5 GPa. Furthermore, the results of RSD-6C align with those reported by Del Rey et al. and others [87–89] for the elasticity of cement-treated CDWs.This implies that the long-term elasticity is influenced by the curing time and the particle size. The increase in RSD-6C elasticity demonstrates its ability to withstand stress and maintain its rehabilitated structure over time. The comparison of SD-6C and RSD-6C highlights the similarities in their response to the curing process, but also reveals differences in their rate of increase in elasticity, making it possible to assess the road material suitability for certain applications after being recycled and treated for the second time.

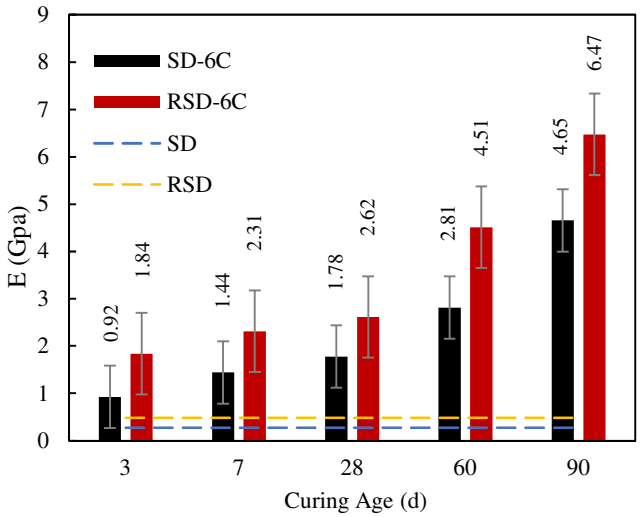


Fig. 6 Modulus of Elasticity of SD, RSD, SD-6C, and RSD-6C at the different curing ages

3.3.4. UCS Correlation with E and ITS

The relationships between the E and UCS and between ITS and UCS for the uncemented materials SD and RSD and along the different curing ages for the cemented materials SD-6C and RSD-6C are shown in **Fig. 7 (a)** and **Fig. 7 (b),** respectively.

Strong correlations between E and UCS (R^2=0.74) and between ITS and UCS (R^2=0.94) were observed. The relationship between E and UCS is consistent with the findings of Tan et al. 2002 [90], which investigated the improvement of marine clay properties through the addition of cement. However, it differs from the results of Ho et al. 2021 study [91], which studied cement-stabilized marine deposits. In contrast, the ITS and UCS correlation results are consistent with the studies of Zentar et al. 2021[63], Hongwei et al. 2022 [84], Wang et al.2018 [92], and Yoobanpot et al. 2020[93], which focused on sediments treated with single or composite binders based on OPC, calcium sulfoaluminate cement, or fly ash, with varying percentages and different constituent ratios.

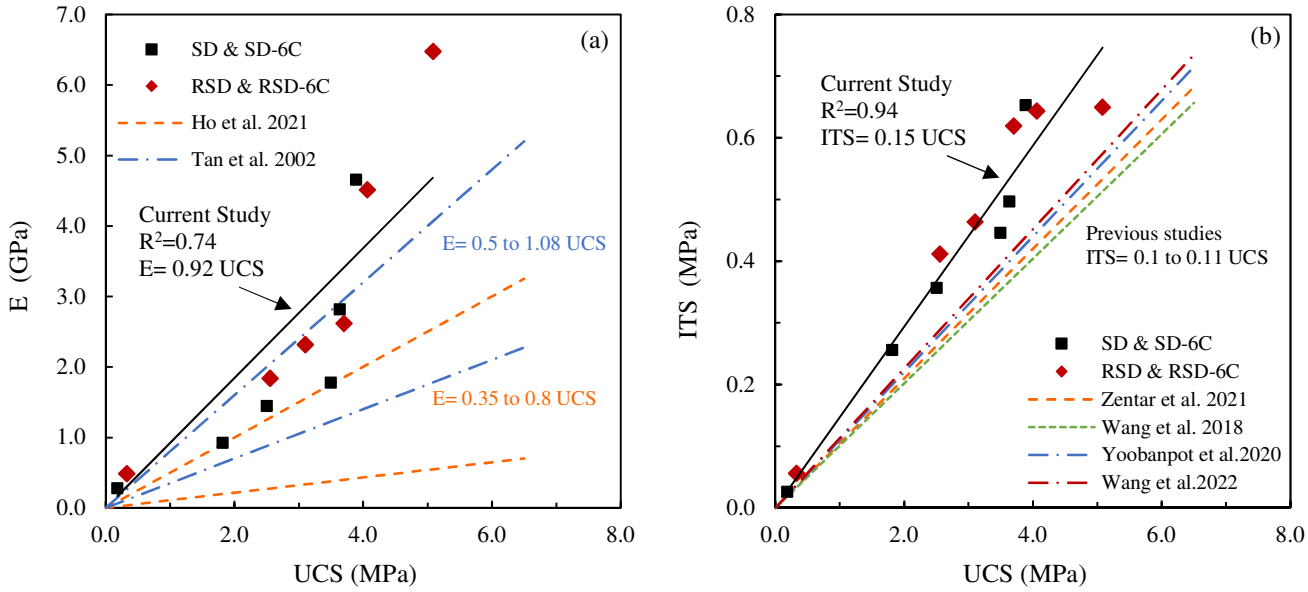


Fig. 7 Correlation of unconfined compressive strength with (a) modulus of elasticity and (b) indirect tensile strength along SD, SD-6C, RSD, and RSD-6C

3.4. Durability Tests

3.4.1. Water sensitivity

Material resistance to water is examined by measuring its retained strength after the material immersion in water [94]. Results are shown in **Fig. 8**. The UCS values of SD-6C and RSD-6C at 60 days of normal curing (60N) were equal to 3.63 MPa and 4.06 MPa respectively. The UCS of specimens subjected to normal curing for 28 days (28N) followed by 32 days of immersion (32I) decreased slightly with the SD specimens displaying a reduction to 3.19 MPa, while the RSD specimens remained unchanged. The water sensitivity ratio (R_w) is defined in equation (2) and it is equal to 0.88 and 1 for SD-6C and RSD-6C respectively.

$$R_W = UCS_{\text{immersed specimens}} / UCS_{\text{control specimens}} \quad (2)$$

Both materials have $R_W > 0.8$ which is higher than the minimum required ratios with respect to their MBV values ($R_w \geq 0.7$ for MBV > 0.5, $R_w \geq 0.8$ for MBV < 0.5) [60]. The MBV is used as an indicator of clay activity and consequently water sensitivity of road materials [95]. The low MBV of SD and RSD indicates their less sensitivity to water. They are less likely to exhibit durability problems, such as cracking and stripping, under various moisture conditions as they have a high resistance ratio.

Similarly, Wang in 2011 [64] reported R_w values of 0.96 and 1.16 for SD with an MBV of 3.1g/100g stabilized with 6% and 9% of cement, respectively. However, Hossain and Mol [96] reported R_w

values greater than 0.8 for clayey soil stabilized with different amounts of cement kiln dust and volcanic ash (5 to 20%), but by involving 91 days of normal curing and 7 days of water immersion.

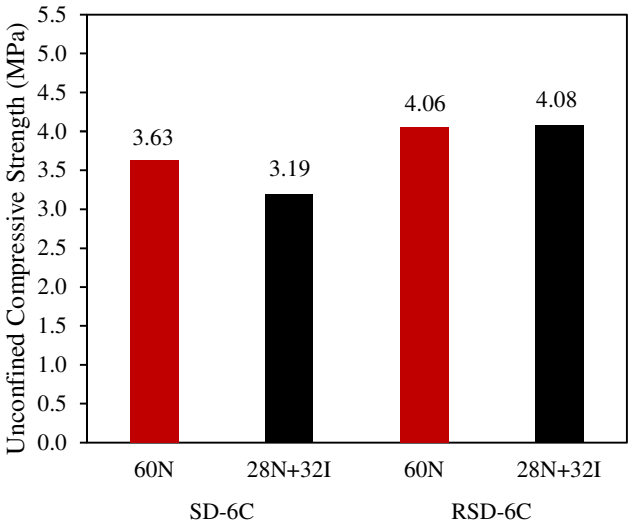


Fig. 8 Water sensitivity of SD-6C and RSD-6C

3.4.2. Freeze-thaw test

The FT process in cold climates results in the penetration of frost into road structures and subgrade soils, causing expansion of pore water and formation of segregation ice. This process, influenced by temperature, moisture, and material susceptibility, leads to increased roughness and cracking of the pavement surface, generation of excessive tensile stresses, and initiation of longitudinal or meandering cracks. Additionally, it causes an increase in the volume of the road's granular structure[97]. To assess the materials' resilience in cold regions as in the north of France, their susceptibility to FT cycles was evaluated using the FT standard NF P98-234-1[55].

In **Fig. 9**, the results of FT tests on the two materials, SD-6C and RSD-6C, are displayed. The materials were compacted using two methods: static axial compression (SC) and vibro-compression (VC). The figure shows the impact of 20 FT cycles on UCS with both compaction types. The percentages in **Fig. 9** represent the strength loss (λ) caused by FT damage, which was calculated using equation (3).

$$\lambda(\%) = \frac{UCS_{\text{Control Specimens}} - UCS_{\text{FT Specimens}}}{UCS_{\text{Control Specimens}}} \times 100 \quad (3)$$

The FT tests resulted in a significant reduction of the UCS values for both materials. Prior to the FT tests, the UCS of SD-6C and RSD-6C was 3.5 MPa and 3.7 MPa under SC and 2.55 MPa and 3.93 MPa under VC, respectively. The difference in UCS values between the two compaction types is due to the different granulometry of SD and RSD. SD, classified as fine soil, was best compacted

using SC, while RSD, classified as sandy material, was best compacted using VC [98]. The compaction method had a significant effect on the UCS values after 20 FT cycles, with the UCS of SD-6C and RSD-6C decreasing to 2.58 MPa and 2.56 MPa under SC, and 1.75 MPa and 3.68 MPa under VC, respectively. The reduction in the UCS values was primarily observed for SD-6C at both SC and VC, with SC showing a reduction of 26.4% and VC a reduction of 31.4%. On the other hand, RSD-6C showed a decrease of 30.8% only in SC and a slight reduction of 6.4% in VC.

Zentar et al. 2012 [61] found similar strength loss values (26% and 29%) for SD stabilized with 6% and 9% of OPC, while SD treated with a composite binder of OPC and fly ash showed a higher strength loss. Additionally, Silitonga et al. 2009 [99] investigated the effects of cement, lime, and two types of fly ashes (Sodeline and Soproline) on dredged sediments. After 7 days of curing, there was a significant decrease in UCS ranging from 49.4% to 74.5%, which was attributed to incomplete hydration and pozzolanic reactions. However, after 28 days of curing, the strength loss decreased to a range of 2.24% to 46.36%, indicating that longer curing times can improve the stabilization results against FT cycles. Moreover, it has been shown that recycled demolition aggregates are more durable to FT cycles than natural aggregates, as reported by Richardson et al. 2011 [100].

These results indicate that SD materials are sensitive to FT cycles, resulting in a substantial loss of strength, unlike RSD which exhibits little change under VC conditions. This behavior is likely due to the fact that fine soils are more susceptible to frost than coarse-grained soils [97]. Furthermore, by taking the observed ratio of strength loss in UCS and applying it to the ITS results as they have a strong correlation(R^2=0.94), it can be inferred that the SD specimens attain a sufficient level of frost resistance (ITS > 0.25MPa) after 28 days of curing, while the RSD specimens exhibit comparable resistance after 7 days of curing.

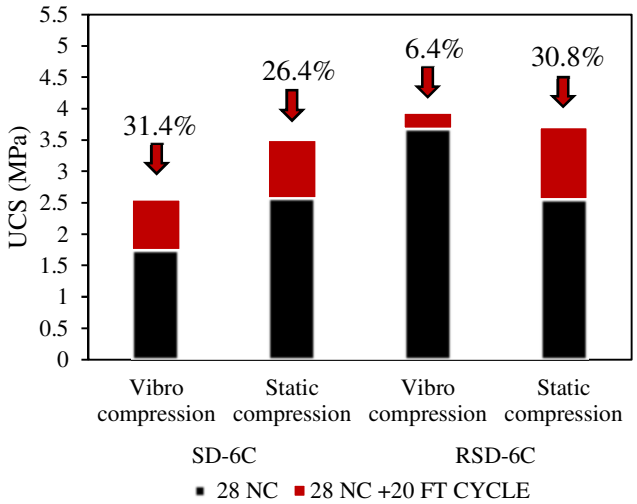


Fig. 9 Freeze-Thaw effect on SD and RSD

3.4.3. Treatment Suitability

The suitability test results indicate that RSD-6C has a higher VS of 1%, compared to SD-6C, which has a VS of 0.51%. RSD-6C also exhibits higher ITS with a value of 0.56 MPa, while SD-6C has a lower ITS of 0.32 MPa. These higher VS and ITS found for RSD-6C are attributed to its coarser granulometry. Dubois et al. 2009 [10] found that a compacted mix of SD, dredged sand, and cement had a lower swelling value of 0.03% compared to SD alone, which had a swelling value of 2%. This can be attributed to the lower swelling properties of the coarser material, dredged sand. However, in the current study, the RSD has more pores (**Table 2**) and consequently more space for absorption of moisture compared to SD, which was observed in its higher OWC. It behaves more like CDW materials that absorb more water [101], resulting in a higher swelling value. Moreover, it tends to have stronger interparticulate bonds due to the old hydrated paste in its matrix. This help to distribute tensile forces evenly throughout the material compared to SD, which results in higher ITS (**Fig. 10**). Both treatments are classified as suitable treatments (VS < 5%, ITS > 0.2MPa) according to materials ITS and VS values upon their immersion under the defined conditions in the standard NF P94-100 [57].

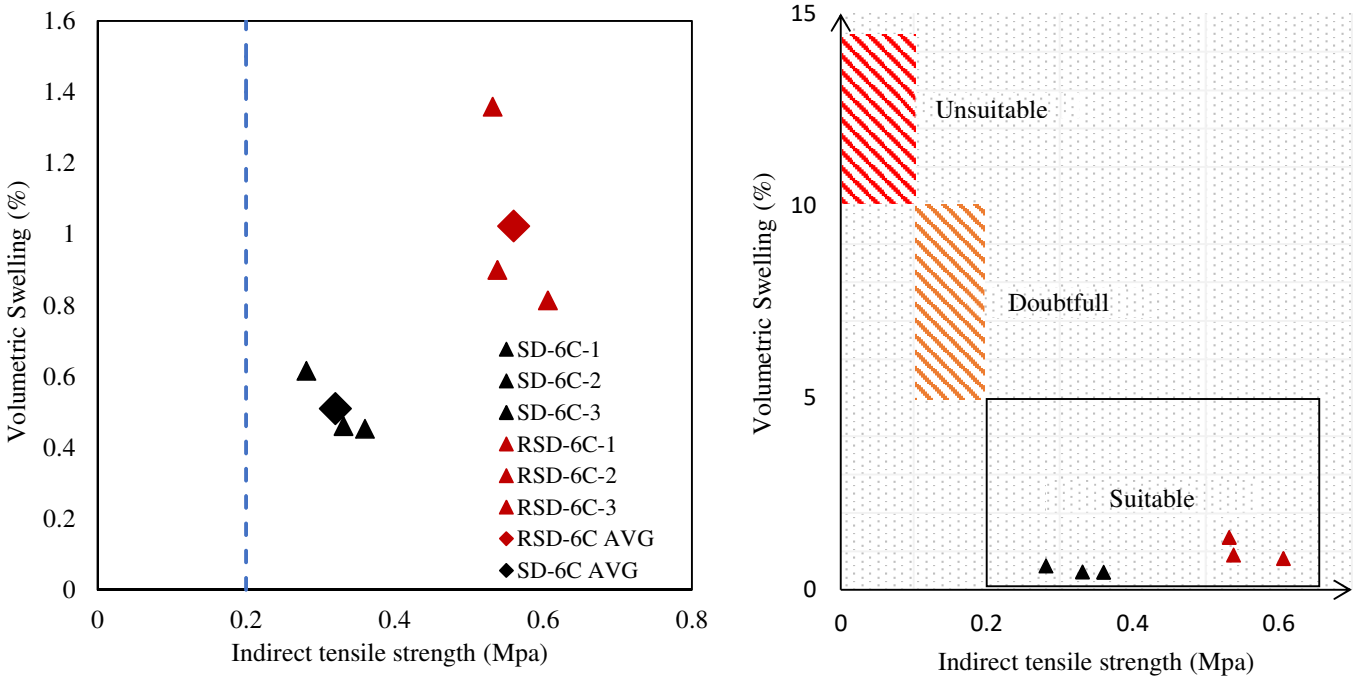


Fig. 10 Treatment Suitability of SD-6C and RSD-6C

3.4.4. Microstructural Analysis

Table 2 and **Fig. 11** represent the microstructural analysis of SD and RSD. They illustrate the total porosity and the differential distribution of porosity along the different pore diameters of the materials SD and RSD. The main pores were found to be distributed in the ranges of 0.01-1.0 and

1.0-40 micrometers(µm). At the initial stage, both SD and RSD showed similar pore structures, within the large pores (1-40µm and > 40 µm) and the smaller pores (<0.01 µm and 0.01-1 µm) in accordance with previous research classification [102].

After 28 days of cement treatment, the porosity shifted towards smaller pore sizes for SD-6C and RSD-6C. This can be attributed to cement hydration, which leads to the transformation of the larger pores into smaller ones, resulting in a finer and denser pore structure. Similar results were obtained by Zentar et al. 2021 [63], where SD treated with 6% of OPC have the highest porosity in the diameter range 0.01-1 µm representing almost 60% of the total porosity after 3 and 7 days of curing. While RSD-6C showed a significant increase in the pore volume percentage of the large pores (> 40 µm) that increase from 4.5% to 8.5%. RSD sandy texture may have affected the behavior of the cement hydration process, leading to the increase in large pore sizes. As a result, the sand-particles-size in RSD may create interconnected large pores that are not filled by the hydration products of the cement, leading to an increase in the pore volume percentage of the large pores. Additionally, RSD particles may also affect the water distribution water in the material, influencing the rate and extent of cement hydration in some areas.

The total porosity results obtained for SD and SD-6C reveal lower values of 29.97% and 27.30%, respectively, compared to the results obtained for RSD and RSD-6C, which show higher values of 32.32% and 30.51%, respectively. Therefore, the behavior of cement hydration process and changes in porosity are dependent on material properties like grain size, mineral composition, and texture.

Table 2 Total-porosity and pore size distribution of SD and RSD

Pore size (µm)	**Pore size distribution (%)**			
	SD		RSD	
	Raw	6C-28d	Raw	6C-28d
> 40	6.7	0.9	4.5	8.5
1-40	44.5	34.0	36.4	28.0
0.01-1	47.4	61.4	56.2	60.3
< 0.01	1.4	3.7	2.9	3.2
Total	100.0	100.0	100.0	100.0
Porosity (%)	29.97	27.30	32.32	30.51

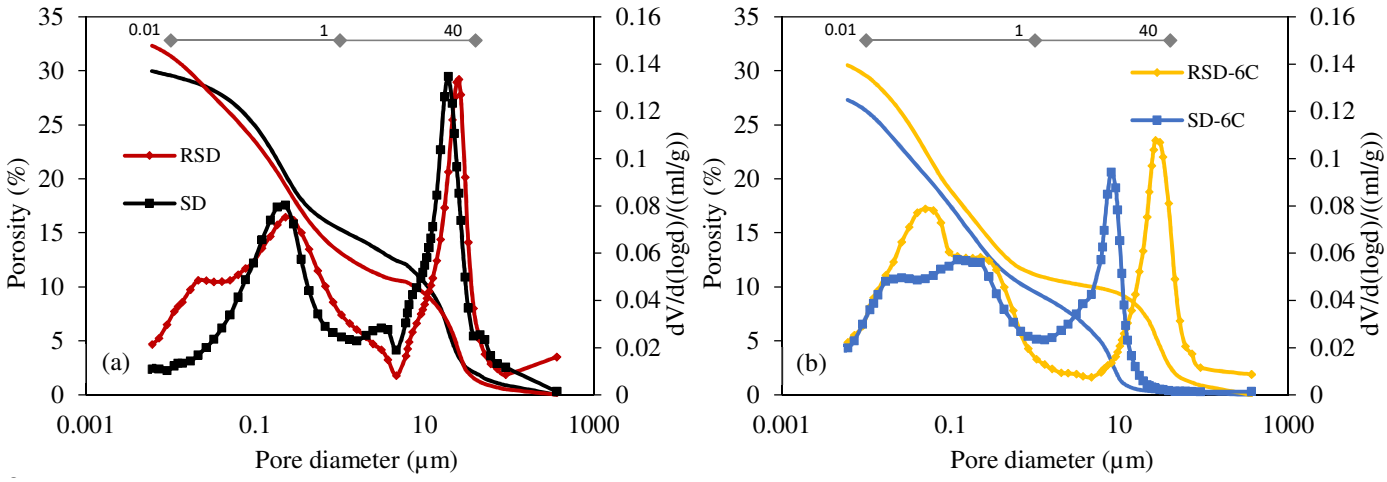


Fig. 11 Microstructural Analysis of (a) Initial materials SD and RSD, and (b) treated materials SD-6C and RSD-6C.

3.4.5. Long-term Performance & Material Classification

The selection of suitable materials for road sub-base and base course is crucial to ensure long-lasting and durable roads. One important criterion for these materials is their ability to withstand traffic loads without failure. Thus, the materials should withstand the stress of vehicular traffic, which demands high tensile strength and the ability to undergo deformation without rupturing. As such, the materials used in these layers must meet the minimum tensile strength (TS) and E requirements to provide optimal structural integrity and durability for roads. TS and E for SD-6C and RSD-6C at 360 days are estimated based on measurements taken at 28 days, using equations (4), (5), and (6) [103] that have been validated by multiple sediments valorization studies [10,13,18] and with respect to the hydraulically bound mixtures standard NF EN 14227-15 classifications [54].

$$TS = 0.8 \times ITS \quad (4)$$

$$TS_{360d} = TS_{28d} / 0.6 \quad (5)$$

$$E_{360d} = E_{28d} / 0.65 \quad (6)$$

Based on **Fig. 12**, the materials SD-6C and RSD-6C are classified as T3 materials, with SD-6C situated precisely on the T3 border and RSD-6C placed in a higher area close to the T4 class. Similar results were reported for solidified sediments by Wang et al. and others [19,63]. As a result, RSD-6C displays a slight improvement in its classification level. Accordingly, the French technical guide [60] recommends the use of RSD-6C in the sub-base layer for the traffic classes T5 up toT1, and it may also be employed in the base course layer for traffic classes T5, T4, and T3, and potentially for T2 or T1, subject to a specific investigation. SD-6C, on the other hand, lies on the boundary between material classes T2 and T3, making it appropriate for use in the sub-base layer for traffic classes T5 to T2, and perhaps T1. However, it can only be used in the base course for traffic classes T5 and possibly T4 or T3.

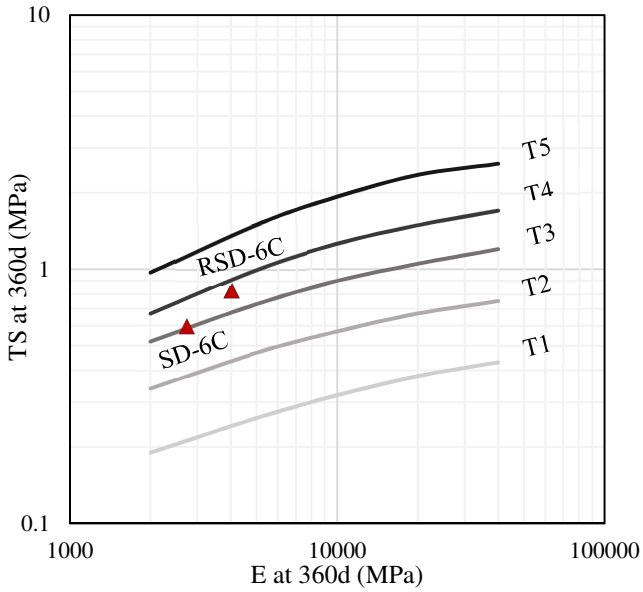


Fig. 12 Road material classification

3.5. Environmental Assessment

SD and RSD are considered non-inert and non-hazardous sediments, as per the French and European classifications [104,105]. These sediments can be evaluated for their potential use in road construction. A methodological guide is available for the acceptance of alternative materials in road applications [16], with recent technical guides released for different types of waste, including CDW[106]. These guides provide specific thresholds of material leaching to be used for covered embankments, paved and unpaved road construction. As a sediment application guide is not yet available, and as RSD falls under the category of CDW waste material, the CDW guide is utilized to evaluate leaching data for both SD and RSD under their initial and treated stages.

The investigation of leaching concentrations of various elements from SD and RSD, before and after treatment, at different curing ages (28, 60, 90) is summarized in **Fig. 13** . Different studies have investigated the environmental impact of solidification/stabilization processes on the leaching behavior of sediments' contaminates [13,62]. However, their evaluation as road materials has been limited to their general acceptability for road applications [107], using the SETRA 2011 methodological guide [16]. In contrast, the current study provides an assessment of sediment contamination specifically for their use in different types of usage in road construction (type1 (t1), type2 (t2), and type3 (t3)).

It was observed that the leaching concentrations of As, Cd, Sb, Se, and Pb were all below the QL of the materials, except for a slightly higher Pb concentration in RSD-6C at 28 and 60 days of curing. However, all concentrations were below the CDW leaching thresholds of the three types of roads t1,

t2, and t3. The concentrations of Ba, Mo, Zn, and chlorides were also found to be below the thresholds for all materials.

In soils containing sulfates, hydraulic binders, such as cement, are frequently used for stabilization [108]. Regarding sulfate leaching release, only RSD-6C at its different curing ages did not exceed the thresholds, in contrast, SD, RSD, and SD-6C all exceeded the t3 sulfate level (1300mg/kg). The cumulative effect of calcium sulfate precipitation in RSD-6C prevented the release of sulfates above the thresholds. The leaching concentrations of Cr and Cu for SD and RSD were below the thresholds. But for the other materials namely SD-6C and RSD-6C, their concentrations increased above t3 level, although they remained below t2 and t1 levels. For Ni, leaching concentrations for SD, SD-6C, and RSD were all below the three thresholds (0.5 mg/kg), however, RSD-6C exceeded these levels at various curing ages. Additionally, only SD met the t3 thresholds for fluoride release, while other materials exceed t2 and t1 at different curing ages. Revalorization and addition of cement affect the solubility of contaminants at high pH, leading to increased leaching concentrations in some pollutants. These findings have important implications for the safe reuse of these materials in road construction.

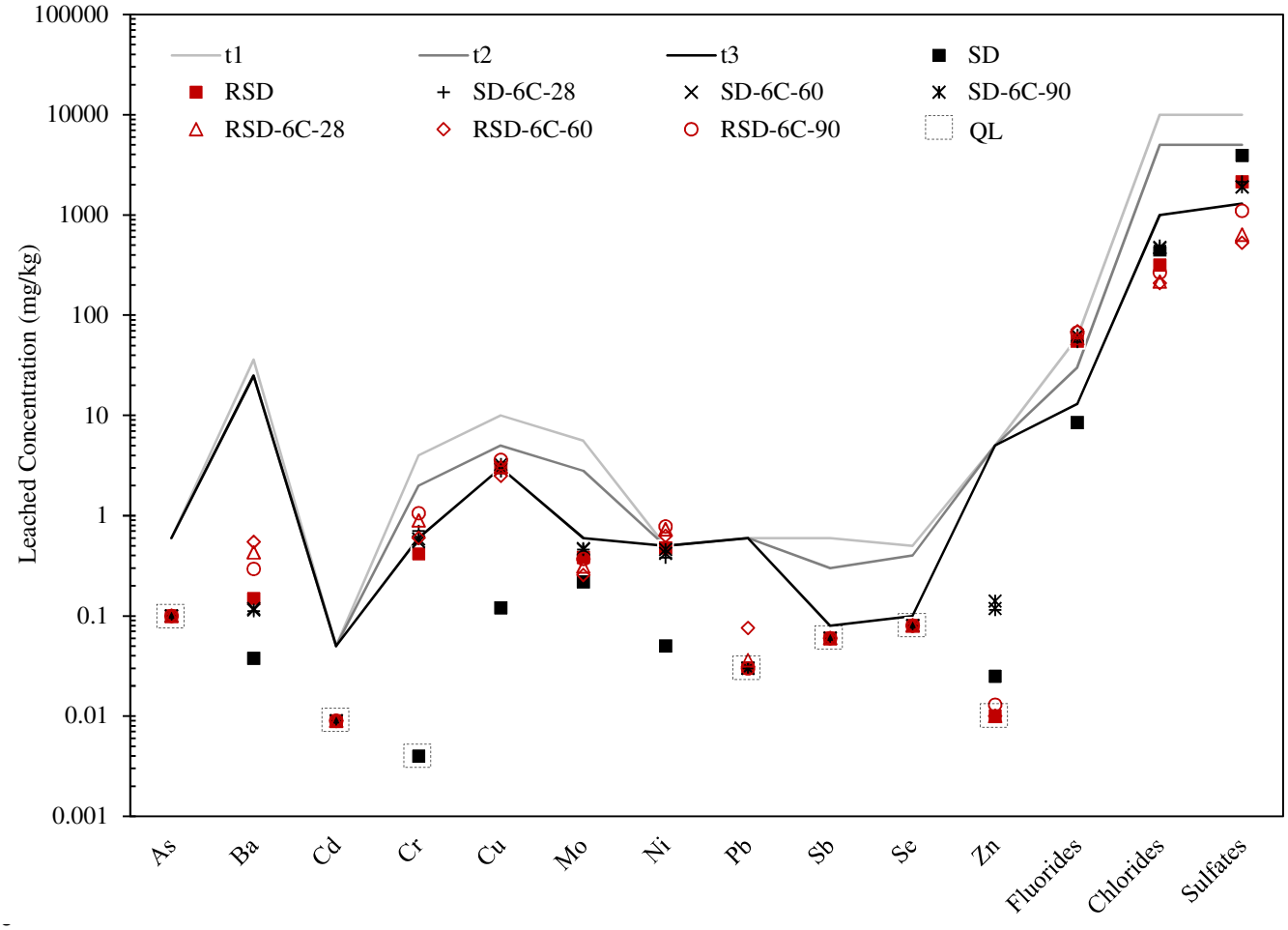


Fig. 13 Leaching concentration of SD, RSD, SD-6C, and RSD-6

4. Conclusion and perspectives

Road construction can benefit from the use of waste materials, like dredged sediments, as an environmentally responsible way to manage resources and decrease reliance on new materials. However, the successful implementation of circular economy principles in road construction requires considering the entire life cycle of these materials, including their end-of-life management and potential for revalorization. This study evaluated the circularity and durability of demolished road materials made from dredged sediments for use in new road structures after road failure, comparing the valorization of raw sediments versus recycled sediments in terms of their mechanical and environmental performance. The results can be summarized as follows:

- The physicochemical evolution mainly demonstrated that RSD has no plasticity, less organic matter content, and lower particle density (2.55 g/cm^3) than SD (2.59 g/cm^3) due to the stabilization effect and the content of the recycled cement paste in the RSD matrix.
- Materials' compaction tests showed that RSD has a lower MDD of 1.7g/cm^3 than SD of 1.79g/cm^3, and both materials have a higher MDD after treatment with 6% of cement.
- The short mechanical analysis revealed that RSD-6C has an I-CBR value of 90% while SD-6C I-CBR is equal to 55%, and both materials have higher I-CBR index than the required for subbase and base course layer. Additionally, along the different curing ages 3,7,28,60, and 90 days, RSD-6C exhibits a higher UCS, ITS, and E than that of SD-6C and consequently higher road material classification. These changes are mainly attributed to the RSD granular composition and lower organic matter due to the recycling effect.
- RSD-6C shows slightly higher volumetric swelling, but its water sensitivity (R_w =1) is less than that of SD-6C (R_w =0.88). Additionally, RSD-6C exhibits greater freeze-thaw resistance in the VC method and attains sufficient frost resistance in a shorter curing period.
- The environmental evaluation of RSD and SD has revealed distinct behavior patterns exhibited by these materials before and after undergoing treatment. This highlights the significance of placing special emphasis on the treating binder, as well as the specified road usage requirements.

Therefore, the results demonstrated the potential of recycled sediments for use in higher-performance road layers, with higher mechanical and durability properties meeting the environmental thresholds for different road usages. These results are significant for the development of sustainable infrastructure and will be valuable to policymakers, engineers, and researchers. Furthermore, this research highlights the importance of conducting studies on the long-term management of valorized materials to ensure their integration into the circular economy chain. By considering revalorization in the design phase, road construction can elevate a more durable and circular approach that reduces waste and advances circular economy principles.
Further research is needed to fully understand the environmental benefits and drawbacks of sediment use on a case-by-case basis and to ensure its environmentally sustainable utilization through a life

cycle assessment that compares revalorization versus landfill scenarios to determine the most optimal end-of-life management scenarios.

Table of Abbreviations

CDW: Construction and Demolition Waste	RSD: Recycled Sediments
E: Modulus of Elasticity	R_w: Water Sensitivity Ratio
FT: Freeze-Thaw	SC: Static Compression
I-CBR: Immediate California bearing Ratio	SD: Raw Sediments
ITS: Indirect Tensile Strength	TS: Tensile Strength
MBV: Methylene Blue Value	UCS: Unconfined Compressive Strength
MDD: Maximum Dry Density	VC: Vibro Compression
OWC: Optimum Water Content	VS: Volumetric Swelling

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Disclosure statement

We certify that there is no actual or potential conflict of interest in relation to this article.

Data availability statement

The authors confirm that the data supporting this study's findings are available within the article. Any other supportable data can be available from the corresponding author [MA], upon reasonable request.

Author Contributions

Mohammad Almokdad: Methodology, investigation, formal analysis, writing - original draft.

Rachid Zentar: Original idea of re-recycling sediments, supervision, resources, re-writing when necessary, and review & editing.

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